MAHEAK DAVE

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OBJECTIVE

AI researcher with 2 years of research experience in deep learning and computer vision, seeking full-time Deep-learning R&D roles.

EDUCATION

Bachelor of Computer Science with specialization in A.I., Techno India University — 2021 - 2025
CGPA - *8.52*

SKILLS

Technical Skills	PyTorch, ONNX,Python, FastAPI, C++, MySQL, MongoDB, ChromaDB, Docker, Git
Soft Skills	Collaboration, Problem-solving, Communication, Time management, Adaptive

EXPERIENCE

Research Intern — Sept 2024 - Sept 2025
DRDO (Defence Research and Development Organization) — *Kolkata, WB*

- Pioneered the paper **Self Distillation via Iterative Constructive Perturbations**, proposing novel constructive perturbation technique and developed an innovative self-distillation strategy, achieving a remarkable 20% relative improvement in neural network performance over baseline metrics. Submitted findings at **ICCV-2025**.
- Lead developer for the DRDO XCA desktop app: architected and implemented a production-ready, unified explainability suite from scratch in PyTorch for both signal and image models (Standard CAM(Class Activation Maps)-variants, I-CAM, HiLite for SVM, LIME, SHAP, guided-backprop variants, and more).
- Integrated in-app image-mask annotation for user validation and a secure custom pretrained-model upload pipeline, enabling researchers to run, inspect and validate explanations on their own models.
- Surveyed on several defences such as gradient obfuscation, Certified radius based, Parceval networks, etc. against image perturbation based adversarial attacks on neural networks, which counters atleast 90% of the attacks.
- Implemented an efficient vectorized pipeline for a novel kernel filtering and selection method from **millions** of image kernels, for directional morphological operations along with road detection using a novel YOLO-v8 based custom implemented FPN (Feature Pyramid Network) module.

Student Researcher — Apr 2023 - Feb 2024
Remote — *Kolkata, WB*

- Published 6 research papers under different specialized university professors available in here - **Google scholar profile**.
- Awarded **best paper award** for a paper submission in a local conference (ICSSAI - 2023).

PROJECTS

RAG from scratch

- Reimplemented the GPT-2 (124M) architecture in PyTorch and loaded pretrained weights via safetensors, enabling full forward/inference functionality with custom sampling and Byte-Pair Encoding tokenization.
- Developed a complete RAG pipeline using FastAPI, ChromaDB, and Chainlit, supporting document ingestion, chunking, embedding generation, semantic retrieval, and contextual prompt assembly.
- Packaged the entire system, LLM server, vector store, API layer, and UI using Docker Compose, delivering a reproducible, end-to-end retrieval-augmented generation stack as a solo developer.